equations are $y_1 = A \sin(\omega t + \varphi_1)$ and $y_2 = A \sin(\omega t + \varphi_2)$, then the phase difference $\Delta\varphi = (\omega t + \varphi_2) - (\omega t + \varphi_1) = \varphi_2 - \varphi_1$.

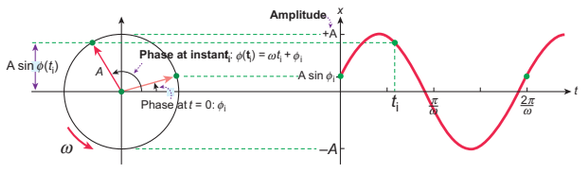


Figure 10.11 The phase of vibrating particle at two instant of time.

EXAMPLE 10.5

A nurse measured the average heart beats of a patient and reported to the doctor in terms of time period as 0.8 *s*. Express the heart beat of the patient in terms of number of beats measured per minute.

Solution

Let the number of heart beats measured be *f*. Since the time period is inversely proportional to the heart beat, then

$$f = \frac{1}{T} = \frac{1}{0.8} = 1.25 s^{-1}$$

One minute is 60 second,

$$(1 \text{ second} = \frac{1}{60} \text{ minute} \Rightarrow 1 \text{ s}^{-1} = 60 \text{ min}^{-1})$$

$f = 1.25 \text{ s}^{-1} \Rightarrow f = 1.25 \times 60 \text{ min}^{-1} = 75$ beats per minute

EXAMPLE 10.6

Calculate the amplitude, angular frequency, frequency, time period and initial phase for the simple harmonic oscillation given below

a. $y = 0.3 \sin(40\pi t + 1.1)$

b. $y = 2 \cos(\pi t)$

c. $y = 3 \sin(2\pi t - 1.5)$

Solution

Simple harmonic oscillation equation is $y = A \sin(\omega t + \varphi_0)$ or $y = A \cos(\omega t + \varphi_0)$

a. For the wave, $y = 0.3 \sin(40\pi t + 1.1)$

Amplitude is $A = 0.3$ unit

Angular frequency $\omega = 40\pi$ rad s^{-1}

Frequency $f = \frac{\omega}{2\pi} = \frac{40\pi}{2\pi} = 20 Hz$

Time period $T = \frac{1}{f} = \frac{1}{20} = 0.05 s$

Initial phase is $\varphi_0 = 1.1$ rad

b. For the wave, $y = 2 \cos(\pi t)$

Amplitude is $A = 2$ unit

Angular frequency $\omega = \pi$ rad s^{-1}

Frequency $f = \frac{\omega}{2\pi} = \frac{\pi}{2\pi} = 0.5 Hz$

Time period $T = \frac{1}{f} = \frac{1}{0.5} = 2 s$

Initial phase is $\varphi_0 = 0$ rad

c. For the wave, $y = 3 \sin(2\pi t + 1.5)$

Amplitude is $A = 3$ unit

Angular frequency $\omega = 2\pi$ rad s^{-1}

Frequency $f = \frac{\omega}{2\pi} = \frac{2\pi}{2\pi} = 1 Hz$

Time period $T = \frac{1}{f} = \frac{1}{1} = 1 s$

Initial phase is $\varphi_0 = 1.5$ rad

EXAMPLE 10.7

Show that for a simple harmonic motion, the phase difference between

a. displacement and velocity is $\frac{\pi}{2}$ radian or 90°.

b. velocity and acceleration is $\frac{\pi}{2}$ radian or 90°.

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